

This is **not a generic stock simulator**—it's a **fast-paced, information-driven trading game built around real stocks, fictional scenario-based news, scarcity, negotiation, and an unexpected second stock release.**

BULLS & BEARS — COMPLETE SIMULATOR SPECIFICATION

1. Core Concept

Bulls & Bears is a real-time multiplayer stock-market simulation operated through **Discord.**

Players start with **\$100,000 in virtual cash** and trade a selection of **real publicly traded stocks.**

The game consists of approximately **8 rounds.**

Each round:

1. A market news flash is released.
2. The news contains **5–6 separate developments.**
3. Each development affects specific sectors/stocks.
4. The actual consequences are **not explicitly stated.**
5. Players analyze the information and decide what to buy/sell.
6. Trading opens for **exactly 10 minutes.**
7. Players can trade with the market or negotiate swaps with other players.
8. Trading closes.
9. Stock prices are updated based on the round's events.
10. The next round begins.

The objective is to finish with the highest portfolio value after the final liquidation.

2. Real Stocks, Fictional News

This is an important distinction.

Stocks = REAL

Use actual publicly traded companies, for example:

- Apple
- Microsoft

- Nvidia
- Tesla
- Amazon
- JPMorgan
- ExxonMobil
- Coca-Cola
- Boeing
- Walmart
- etc.

The initial universe can contain approximately **50 stocks**.

News = FICTIONAL

The news should **not claim to be actual breaking news**.

Instead, it should be based on **plausible real-world scenarios**.

For example:

GLOBAL MARKETS — 11:00 AM

The European Union announces a hypothetical accelerated transition toward electric commercial fleets, with major incentives for battery-powered transportation.

The event is fictional, but the scenario is realistic enough that players can reason about which companies might benefit.

The simulator should clearly state that all news is fictional and created for the game.

3. News Flash Design

This is one of the most important mechanics.

Every news flash contains approximately:

5–6 separate news stories

Each story should affect approximately:

2–3 sectors/stocks

However, **do not tell players what the consequences are**.

BAD

Apple will benefit significantly from increased demand.

That's giving away the answer.

GOOD

A major Southeast Asian electronics manufacturer announces a hypothetical multi-year expansion of its semiconductor packaging capacity.

The players need to figure out:

Who benefits?

Who gets hurt?

Which companies are exposed?

Is this immediate or does it affect future rounds?

4. News Should Contain Clues

The news shouldn't simply say:

Company X has a problem.

It should provide enough information for intelligent players to make deductions.

For example:

Fictional News

Several European automakers announce plans to accelerate EV production over the next three years. Battery manufacturers report that securing lithium supply has become increasingly difficult.

Players might infer:

EV → batteries → lithium → mining → supply chain

But the simulator should never explicitly say:

"Buy Nvidia."

or

"Tesla will rise."

The player has to figure it out.

5. Multiple Sectors Per News Flash

A single news flash should create a **mixed market**.

For example:

NEWS FLASH — ROUND 3

Story 1 — Technology

A hypothetical regulatory framework changes the economics of large-scale AI infrastructure.

Potentially relevant:

- Nvidia
- Microsoft
- Amazon

Story 2 — Energy

A hypothetical geopolitical disruption threatens a major oil-producing region.

Potentially relevant:

- ExxonMobil
- Chevron
- Airlines

Story 3 — Consumer

A major global retailer announces an unexpected change in its international expansion strategy.

Potentially relevant:

- Walmart
- Nike
- logistics companies

Story 4 — Financials

A hypothetical central-bank policy change alters the interest-rate environment.

Potentially relevant:

- JPMorgan
- Goldman Sachs
- real-estate-related companies

Story 5 — Healthcare

A hypothetical regulatory decision changes the approval pathway for a class of drugs.

Potentially relevant:

- pharmaceutical companies

Story 6 — Industrial

A hypothetical infrastructure spending package accelerates demand for construction equipment.

Potentially relevant:

- Caterpillar
- industrial suppliers

Again, **the game never tells players the resulting price movement beforehand.**

6. Stock Universe

Initial Market

Approximately:

50 real stocks

Each stock has:

- Ticker
- Company name
- Sector
- Starting price
- Maximum market supply
- Current market price
- Available shares

Example:

Stock	Sector	Starting Price	Market Supply
NVDA	Technology	\$180	100
AAPL	Technology	\$250	100

Stock	Sector	Starting Price	Market Supply
TSLA	Automotive	\$420	100
JPM	Financials	\$300	100
XOM	Energy	\$145	100

The exact starting prices can be configured by the administrator.

7. 100-Share Market Limit

Each stock has a maximum of approximately:

100 shares available in the market

This creates scarcity.

Example:

NVDA

Market supply:

100 shares

Players collectively buy:

80 shares

Remaining market supply:

20 shares

If another player wants 30 shares, they **cannot buy 30 from the market**.

They could:

- Buy the available 20
- Find another player holding NVDA
- Negotiate a swap

This is what makes the player-to-player market important.

8. Starting Capital

Every participant starts with:

\$100,000 virtual cash

No real money.

Starting portfolio:

\$100,000

Starting holdings:

0 shares

9. Buying From the Market

Players can buy stocks directly from the market.

Example:

/buy NVDA 20

The system checks:

- Is NVDA available?
- Are enough shares available?
- Does the player have enough cash?
- Is trading currently open?

If valid:

 BUY EXECUTED

NVDA × 20

Price: \$180

Cost: \$3,600

10. Selling To the Market

Players can sell shares back to the market.

/sell NVDA 10

The system checks:

- Does the player own 10 NVDA?
- Is trading open?

Then:

✅ SELL EXECUTED

NVDA × 10

Price: \$210

Proceeds: \$2,100

The shares return to the market's available supply.

11. Player-to-Player Trading

This is a **major part of the game**.

Players can physically negotiate with each other.

Example:

Trader A

"I'll give you 20 Nvidia if you give me 30 Tesla."

They negotiate **in person**.

Once they agree, they execute the transaction through Discord.

12. NO AUTOMATIC VALUATION OF SWAPS

This is important.

The simulator should **not determine whether a swap is financially fair**.

It should not say:

"20 NVDA is worth \$X, therefore you must give Y."

The market price is simply information.

The players decide what they are willing to exchange.

This creates actual negotiation.

13. Keep Swaps Simple

Do **not** support:

- Stock + cash

- Multiple stocks on one side
- Complex multi-party trades
- Complicated derivatives

Keep it simple:

Stock A × quantity ↔ Stock B × quantity

Example:

/swap @TraderB NVDA 20 TSLA 15

Trader A gives:

20 NVDA

Trader B gives:

15 TSLA

Both players must confirm.

14. Swap Confirmation

Example:

Trader A

 TRADE PROPOSAL

You give:

20 NVDA

You receive:

15 TSLA

Waiting for Trader B.

Trader B

@TraderB has proposed a trade.

Give: **15 TSLA**

Receive: **20 NVDA**

/confirm

Once both confirm:

✅ SWAP EXECUTED

The system automatically updates both portfolios.

15. Trading Window

This needs to be strict.

Every round gets exactly:

🕒 **10 MINUTES**

Example:

11:00 — News released

Players read and analyze it.

11:02 — Trading opens

10-minute countdown begins.

11:12 — TRADING CLOSED

No:

- Buys
- Sells
- Swaps

after the deadline.

The system should automatically lock transactions.

16. ROUND STRUCTURE

There should be approximately:

8 ROUNDS

A round follows:

PHASE 1 — NEWS

News flash released.

↓

PHASE 2 — ANALYSIS

Players interpret the news.

↓

PHASE 3 — TRADING

10 minutes

↓

PHASE 4 — CLOSE

Trading automatically shuts down.

↓

PHASE 5 — PRICE UPDATE

The administrator/system applies the consequences.

↓

PHASE 6 — NEXT ROUND

Repeat.

17. ROUND 0 — SETUP

Before Round 1:

Players:

- Join Discord
- Choose their trader name
- Receive \$100,000
- See the initial 50 stocks
- See current prices
- Learn the commands
- Learn how swaps work

No complicated gameplay yet.

This is essentially the onboarding period.

18. ROUNDS 1–4

The first four rounds operate with the initial:

50-stock market

Each round introduces a new collection of fictional news.

The events should gradually become more interconnected.

Players should start learning:

- Which companies are exposed to which sectors
 - How other traders behave
 - Which stocks are becoming scarce
 - How to negotiate
 - How to anticipate future events
-

19. THE ROUND 4 TWIST

This is one of the coolest mechanics.

At the end of Round 4, announce:

MARKET EXPANSION

25 NEW STOCKS HAVE ENTERED THE MARKET.

The market goes from:

50 stocks → 75 stocks

The new stocks should have their own:

- Current price
 - Sector
 - 100-share market supply
-

20. Why the 25 New Stocks Matter

This should create a major strategic disruption.

Some players may have spent almost all of their:

\$100,000

on the original stocks.

Now suddenly:

25 new investment opportunities appear.

Those players may have:

\$2,000 cash

while another player may have:

\$40,000 cash

The second player now has substantially more flexibility.

But don't artificially punish players who spent their cash.

The new market should simply create a new strategic situation.

21. New Stocks Should Be Relevant

The 25 new stocks shouldn't just be random additions.

They should introduce **new sectors/companies that become relevant to the upcoming news.**

For example:

Rounds 1–4 focus heavily on:

Technology

Energy

Financials

After Round 4, introduce stocks from:

Healthcare

Industrials

Consumer

Aerospace

Materials

Telecommunications

Then future news can create opportunities around those companies.

This means players have to reconsider their portfolios.

22. ROUNDS 5–7

Now the market contains:

75 stocks

The news becomes more interconnected.

Players now have:

- More stocks
- More scarcity
- More possible trades
- More information to process
- Different amounts of remaining cash

This should make the second half substantially more chaotic.

23. Round 8 — FINAL ROUND

Round 8 is the final trading round.

After the final news event:

Players get their final:

10-minute trading window

Then trading closes permanently.

24. FINAL LIQUIDATION

After Round 8:

FINAL LIQUIDATION

Every player must sell all remaining shares back to the market.

No player can finish with stocks.

Their final score is:

Cash + liquidation proceeds

Since all holdings have been sold, this becomes their final cash balance.

25. WIN CONDITION

The winner is simply:

The player/team with the highest final portfolio value.

Everyone begins with:

\$100,000

Example:

Trader	Final Value
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Trader A	\$182,450
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Trader B	\$164,200
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Trader C	\$151,700
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The system should automatically calculate and announce the final standings.

26. PLAYER PORTFOLIO

Players should be able to use:

/portfolio

Example:

ATLAS — PORTFOLIO

 Cash: **\$32,450**

NVDA — 20 shares

AAPL — 15 shares

XOM — 30 shares

Holdings Value: **\$68,900**

Portfolio Value: \$101,350

P/L: **+\$1,350**

27. MARKET COMMAND

/market

Should show something like:

Ticker Price Available

NVDA \$210 20

AAPL \$245 55

TSLA \$390 0

JPM \$315 72

This lets players see:

- Current prices
 - Market availability
 - Scarcity
-

28. LEADERBOARD

/leaderboard

Could display:

BULLS & BEARS

1. Atlas — \$142,350
2. Nova — \$136,700
3. Orion — \$129,450
4. Zenith — \$121,800

I'd make this configurable by the organizer so they can choose whether it's visible throughout the event or only at certain points.

29. ADMIN SYSTEM

The organizer needs complete control.

Game

/start

/pause

/resume

/end

Rounds

/start-round

/end-round

News

/release-news

Market

/update-price

/update-prices

/add-stocks

Players

/view-player

/view-portfolio

/freeze-player

Leaderboard

/leaderboard

Emergency

/cancel-trade

/adjust-cash

/adjust-shares

30. ADMIN NEWS CONFIGURATION

The organizer should be able to prepare all 8 rounds **before the event**.

For each round:

Round Number

News Flash

News Story 1

News Story 2

News Story 3

News Story 4

News Story 5

News Story 6

Affected stocks/sectors

Price changes

Trading duration

The **affected stocks and price changes** should remain hidden from participants.

Only the news itself is published.

31. PRICE UPDATE SYSTEM

After trading closes, the administrator/system applies the predetermined price movements.

Example:

Before news:

NVDA — \$180

After the round:

NVDA — \$225

The system automatically updates everyone's portfolio value.

Participants should **not see the hidden price-impact mapping before the update**.

32. IMPORTANT — DON'T MAKE NEWS TOO OBVIOUS

The game shouldn't become:

News says X → Buy X.

Instead, players should have to connect multiple pieces of information.

For example:

A hypothetical government announces major investment in domestic semiconductor manufacturing.

This could potentially affect:

- Semiconductor manufacturers
- Equipment suppliers
- Cloud companies
- Energy demand
- Certain chip designers

The player needs to determine the relationships.

33. TRANSACTION LOG

Every transaction needs to be permanently recorded.

Record:

- Timestamp
- Round
- Trader
- Transaction type
- Stock
- Quantity
- Price
- Counterparty
- Confirmation status

Example:

ROUND 4

11:07:32

Trader: Atlas

Action: BUY

Stock: NVDA

Quantity: 20

Price: \$210

Total: \$4,200

For swaps:

ROUND 5

11:04:17

Trader A: Atlas

Trader B: Nova

Atlas → 20 NVDA

Nova → 15 TSLA

Confirmed by both.

This gives organizers a complete audit trail.

34. ANTI-CHEAT

The system should ensure players cannot:

- Buy more shares than available
- Sell shares they don't own
- Spend more cash than they have
- Execute trades after the 10-minute window
- Execute a swap without both parties confirming
- Duplicate transactions
- Modify their own portfolio
- Access hidden price information
- See upcoming news
- See the hidden effects of current news

Administrators should have access to everything.

35. DISCORD EXPERIENCE

The participant experience should feel extremely simple.

They shouldn't have to understand a complicated financial platform.

Their workflow should basically be:

Read news → Think → Talk to other players → Trade → Check portfolio → Repeat

The commands should therefore be extremely intuitive.

Essential commands

/buy [ticker] [quantity]

/sell [ticker] [quantity]

/swap [player] [ticker] [quantity] [ticker] [quantity]

/portfolio

/market

/leaderboard

And the bot should provide clear error messages.

Example:

 You cannot buy 30 NVDA.

Only **12 shares** are currently available in the market.

36. COMPLETE GAME FLOW

BEFORE EVENT

Players join Discord.

↓

Choose trader name.



Receive instructions.



Learn commands.



ROUND 0

\$100,000 distributed.

50 stocks revealed.



ROUND 1

 News Flash



10-minute trading



Market closes



Prices update



ROUND 2

 New News Flash



10-minute trading



Prices update



ROUND 3

 New News Flash

↓

10-minute trading

↓

Prices update

↓

ROUND 4

 New News Flash

↓

10-minute trading

↓

Prices update

↓

 **25 NEW STOCKS RELEASED**

50 → **75 stocks**

↓

ROUND 5

 New News Flash

↓

10-minute trading

↓

Prices update

↓

ROUND 6

 New News Flash

↓

10-minute trading

↓

Prices update

↓

ROUND 7

 New News Flash

↓

10-minute trading

↓

Prices update

↓

ROUND 8 — FINAL

 Final News Flash

↓

10-minute trading

↓

 **FINAL LIQUIDATION**

Everyone sells all remaining stocks.

↓

 **FINAL LEADERBOARD**

Highest final portfolio value wins.

The core philosophy for the developer

I'd put this at the very top of the development brief:

Bulls & Bears should feel like a live financial war of information rather than a conventional stock-trading simulator.

The players should never be explicitly told what to buy. They receive incomplete but meaningful information through fictional real-world scenarios, interpret the implications themselves, compete for scarce shares, negotiate trades with other players, and adapt when the market suddenly expands after Round 4.

The simulator handles the rules and execution; **the players provide the strategy, analysis and negotiation.** That is the part I would protect most closely during development.